

MATH AA HL

Internal Assessment

Finding the most accurate model (amongst the Logistic Growth, SIR, SEIQR Models) in predicting the propagation of COVID-19 in Mauritius?

Rationale

Recently, I came across an article about COVID-19 where I was surprised by the fact that the pandemic was 5 years ago, yet it still felt as though it occurred not long ago. The pandemic rattled the whole world which led to some drastic changes in society. Isolated at home and following online classes completely altered my lifestyle. Eventually, my ears tuned out the persistent headlines pertaining to COVID-19 whilst I began to wonder when things would go back to normal. After reading the article, I became interested in how analysts used mathematics to predict the rate of infection which helped governments to implement the correct measures in time. This is exemplified by the use of different models that gave an idea about the number of people who needed immunity to stop the propagation of the virus.

At first I came across the Logistic Growth Model which is commonly used to depict the infection rate of diseases. The Logistic Growth Model is the simplest model as it consists of a single equation where its best use case is said to be for early-stage predictions before government interventions. However, another model that I found was the SIR Model as it is presumed to have a high accuracy at modelling herd immunity and peak infections which takes the recoveries and deaths into account. After conducting more research, I found a detailed model, the SEIQR Model, where it takes incubation (amount of time for symptoms to develop after being exposed to the virus) periods and recovered individuals into consideration. This piqued my interest, as it showed that different methodologies were used to predict the propagation of a virus during the same time period.

Henceforth, my chosen area of exploration would be to compare the Growth Logistic, SIR, SEIQR Models' accuracies by calculating their own respective percentage error margin pertaining to the real-world context of COVID-19 pandemic in Mauritius. The aim of this IA is to determine **which model (amongst the Logistic Growth, SIR, SEIQR Models) is the most accurate in predicting the propagation of COVID-19 in Mauritius?**

Introduction

Context of Exploration

The Logistic Growth Model is the simplest model used widely to describe epidemics as it shows how they grow exponentially at first and after a time period reach a maximum limit. The SIR Model is an epidemic model which illustrates predicted trends of a virus propagation in an enclosed population consisting of susceptible, infected and recovered. The SEIQR Model is another epidemiological model constructed to replicate the transmission of contagious diseases (COVID-19) where a country's population comprises those who are susceptible, exposed, infected, quarantined and recovered.

Defining Key Terms and Variables

The independent variable will be assessed as the number of days (t) since the model depicts how the pandemic spreads based on how long it persists in the population.

The interconnected dependent variables of all 3 models are defined as follows:

- $S(t)$ - **Susceptible population (SIR & SEIQR models)** - individuals who have not contracted the virus, however they are prone to the virus.
- $E(t)$ - **Exposed population (SEIQR Model only)** - individuals who are infected but not yet contagious, this is during the latent phase.
- $I(t)$ - **Infected population (All models)** - confirmed individuals who have contracted the virus and are now sick.
- $Q(t)$ - **Quarantined population (SEIQR Model only)** - individuals who are hospitalised or isolated from the general population and are assumed to have contracted the virus.
- $R(t)$ - **Recovered population (SIR & SEIQR models)** - individuals who have recovered and are now immune to the virus.

The correlation between these variables will form the premise of the aforesaid models which will be derived later on.

Assumptions

The SIR and SEIQR models make the same assumptions:

1. Every person in the population has an equal probability of interacting with any other person since the population is assumed to interact randomly. Hence, not taking geographical variations and societal structures into consideration.
2. Pertaining to the recovered population, the models assume that the latter cannot be reinfected again and are immune to the disease, thus ignoring reinfection and reduced immunity over time.

The Logistic Growth and SEIQR models draw the same assumptions:

Both models presume that cases grow gradually; in Logistic Growth, infections begin in the exponential phase before slowing down, whereas for SEIQR Model, those susceptible enter the $E(t)$ before contracting the virus.

The 3 models draw some assumptions as stated below:

1. Fixed population (N) where there are no net births, deaths and migration.

$$N_{SEIQR} = S(t) + E(t) + I(t) + Q(t) + R(t)$$

$$N_{SIR} = S(t) + I(t) + R(t)$$

$$N_{Logistic} = c \text{ where } c \text{ is a constant}$$

2. It will be assumed that transmission, quarantine and recovery rates are constant over the time period for the SEIQR Model, same as for the transmission rate for the SIR Model and Logistic Growth. Therefore, it does not account for government interventions fully nor alterations in human behaviour which have an impact on those rates where in actuality those rates are always changing due to external factors.

The Logistic Growth Model

Derivation of equation

The Logistic Growth Model illustrates the infected population under resource constraints. It will be impossible for the cases of infection to increase indefinitely, where carrying capacity K , which denotes the maximum population the country can handle, restricts this as the cases will start to increase at a decreasing rate. The logistic equation is as follows:

$$\frac{dI}{dt} = r I \left(1 - \frac{I}{K}\right) \text{ (eq1)}$$

where $I(t)$ is the infected population at time t , r is the growth rate (transmission rate) and $\left(1 - \frac{I}{K}\right)$ limits the growth of the infected population. If $I \rightarrow K$, $\frac{I}{K} \rightarrow 1$, hence $\left(1 - \frac{I}{K}\right) \rightarrow 0$ indicating that the growth begins to slow down until it stabilises at K .

The SIR Model

Derivations of SIR Equations

In order to illustrate how these populations are changing over time, the derivatives of each population with regard to the independent variable (t) are calculated. The following is the derivation of the equations:

Susceptible population

The infection begins when a susceptible individual has contact with an infected person. Hence, the only outflow of the susceptible population is by getting infected. The rate of infection is dependent on the number of susceptible people (S) and a constant rate of infection per contact (β), more infected people (I) will have a higher chance of getting infected. The susceptible population will decline as more people become infected, as indicated by the negative sign. Hence the virus transmission is the rate at which the susceptible population changes at a time period, t .

$$\frac{dS}{dt} = -\beta SI$$

Infected Population

The only inflow will be from the susceptible population at the rate βSI . Its outflow is the recovery of those infected at constant recovery rate (γ). Therefore, the rate of change of the infected population at a given time period, t , shows how infected individuals recover.

$$\frac{dI}{dt} = \beta SI - \gamma I$$

Recovered Population

This population only has an inflow of the infected recovering at the same rate(γ). Those who died from this virus are also part of this population. The rate of change of the recovered population shows how the number of current cases is declining as the latter start to recover.

$$\frac{dR}{dt} = \gamma I$$

The SEIQR Model

Derivations of SEIQR Equations

The derivatives of each population with respect to the independent variable (t) are derived in order to depict the rate of change of these populations over time. The equations are derived as shown below:

Susceptible population

The infection begins when a susceptible individual has contact with an infected person. Hence, the only outflow of the susceptible population is by getting exposed to the disease. The rate of infection is dependent on the number of susceptible people (S) and a constant rate of infection per contact (β); more infected people (I) will have a higher chance of those susceptible getting exposed. Since more people get infected, the susceptible population will decrease which will be connoted by the negative sign. Hence the virus transmission is the rate at which the susceptible population changes at a time period, t .

$$\frac{dS}{dt} = -\beta SI$$

Exposed Population

The inflow of the exposed population will be from the susceptible population at the rate of βSI whilst the first outflow is at a constant rate at which they begin to gain symptoms and become contagious(σ), $E(t) \rightarrow I(t)$. The second outflow is that some exposed people can naturally recover before being contagious at a constant recovery rate(γ_E), $E(t) \rightarrow R(t)$. The third outflow is some

exposed individuals may already be identified early and quarantined before becoming contagious at a constant quarantine rate(δ_E), $E(t) \rightarrow Q(t)$. Thus, the rate at which the exposed population changes at a given time period, t , indicates how those who contracted the virus will enter the latent phase prior to being contagious.

$$\frac{dE}{dt} = \beta SI - \sigma E - \gamma_E E - \delta_E E$$

$$\frac{dE}{dt} = \beta SI - (\sigma + \gamma_E + \delta_E)E$$

Infected Population

The only inflow will be from the exposed population at the rate σE . The first outflow is getting quarantine at rate $(\delta_I) I(t) \rightarrow Q(t)$. The second outflow is the successful treatment of those infected at recovery rate (γ_I) , $I(t) \rightarrow R(t)$. Therefore, the rate of change of the infected population at a given time period, t , shows how infected individuals recover or are quarantined.

$$\frac{dI}{dt} = \sigma E - \gamma_I I - \delta_I I$$

$$\frac{dI}{dt} = \sigma E - (\gamma_I + \delta_I)I$$

Quarantined Population

There are two inflows as shown above are $\delta_E E$ and $\delta_I I$. The only outflow is the recovery of those quarantined with a recovery rate of $\gamma_Q, Q(t) \rightarrow R(t)$. Henceforth, the rate of change of the quarantined population at a given time period, t , indicates how quarantine measures reduce the propagation of the virus.

$$\frac{dQ}{dt} = \delta_E E + \delta_I I - \gamma_Q Q$$

Recovered Population

This population only has inflows namely, $\gamma_E E$, $\gamma_I I$ and $\gamma_Q Q$; people recover from the exposed, infected and quarantine population. The rate of change of the recovered population depicts the reduction in active cases as the latter begin to recover or some have passed away from the virus.

$$\frac{dR}{dt} = \gamma_E E + \gamma_I I + \gamma_Q Q$$

Real-world implementation using COVID-19 data

Real World COVID-19 data

The majority of the data was obtained from OWID and the number of active cases were taken from Worldometer. Due to testing delays, inconsistent reporting, or unexpected outbreaks, daily cases can fluctuate greatly, whilst cumulative cases offer a smoother curve that is less prone to sudden fluctuations and it is easier to model and make comparisons between the 3 models. The time period for the cumulative cases will be taken from 2 Aug 2021(the number of cases began to increase exponentially) to 30 Mar 2025 which is 1336 days (taking initial date as day 0). Referring to the aforesaid assumption, the Mauritian population was fixed at 1.27 million people (3 s.f). However, OWID gives the cumulative confirmed cases per million, thus, the following equation was used to get the total cumulative cases.

$$\text{Total cumulative confirmed cases} = \text{Cumulative confirmed cases per million} \times \frac{\text{population}}{1000000}$$

$$\text{Total cumulative confirmed cases} = \text{Cumulative confirmed cases per million} \times 1.27$$

Root Mean Square Error (RMSE):

The distance between the data points and the regression line is indicated by residuals. RMSE measures the degree of dispersion of these residuals; it shows how concentrated the data is around the line of best fit. In this situation, the RMSE would indicate the number of cases in which a model deviates from the real-world data. Its formula is shown below:

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (P_{model,i} - P_{real,i})^2}{n}}$$

Where,

n : the number of days(1337 days from 2 Aug 2021 to 30 Mar 2025)

$P_{model,i}$: the model's total predicted cumulative cases on day i

$P_{real,i}$: the actual total cumulative cases on day i

A simplified example of the Logistic Growth Model's RMSE is shown below for 3 days after 2 Aug 2021:

$$RMSE = \sqrt{\frac{(4449.46-4471)^2 + (4528.28-4570)^2 + (4608.47-4570)^2}{3}} = \sqrt{\frac{(-21.54)^2 + (-41.72)^2 + (38.47)^2}{3}} = 35.0 \text{ cases (3s.f)}$$

Hence, the complete calculation for the 1337 days for each model is executed in Python Programming Language using the `mean_squared_error` function from `sklearn.metrics` module:

```
35 # Calculating predictions for actual days for RMSE
36 model_fit_to_actual = logistic_fixed_P0_K(real_days, r_fit)
37 rmse = np.sqrt(mean_squared_error(real_cases, model_fit_to_actual))
```

Percentage Error Margin:

The % error margin contextualises the RMSE within the real-world data's scale, thus allowing for a better comparison between the 3 models. Its formula is shown below:

$$\% \text{ Error Margin} = \left(\frac{RMSE}{\text{Mean of Actual Cumulative Cases}} \right) \times 100$$

A simplified example of the Logistic Growth Model's % Error Margin is shown below for the same 3 days after 2 Aug 2021:

$$\% \text{ Error Margin} = \left(\frac{35.045}{\left(\frac{4471+4570+4570}{3} \right)} \right) \times 100 = 0.772\% (3s.f)$$

Thus, the complete calculation for the 1337 days for each model ,using their respective calculated RMSE, is written in Python Programming Language as shown below:

```
39 # Calculating percentage error margin from RMSE
40 mean_actual = np.mean(real_cases)
41 percent_error = (rmse / mean_actual) * 100
```

Logistic Growth Model

Integrating Logistic Growth Equation:

In order to find the infected population at a particular point in time, (eq1) has to be integrated using separation of variables. $I(t) = I$ for simplicity.

$$\frac{dI}{dt} = rI \left(1 - \frac{I}{K} \right)$$

$$\frac{dI}{I(K-I)} = \frac{r}{K} dt$$

$$\int \frac{K}{I(K-I)} dI = \int r dt$$

Using partial fractions, $\frac{K}{I(K-I)} = \frac{A}{I} + \frac{B}{K-I} \Rightarrow A(K-I) + BI = K$ where $A = 1, B = 1$,

$$\int \frac{1}{I} + \frac{1}{K-I} dI = \int r dt$$

$$\ln \left| \frac{I}{K-I} \right| = rt + c \text{ where } c \text{ is a constant}$$

$$\frac{I}{K-I} = e^{rt+c}$$

Let $e^c = Y$ where Y is a constant,

$$\frac{I}{K-I} = Ye^{rt} \quad (eq2)$$

$$I = Ye^{rt}(K - I)$$

$$I = YKe^{rt} - YIe^{rt}$$

$$I + YIe^{rt} = YKe^{rt}$$

$$I(1 + Ye^{rt}) = YKe^{rt}$$

$$I(t) = \frac{YKe^{rt}}{1 + Ye^{rt}} \quad (eq3)$$

From (eq2), when $t = 0$, $I = I_0$,

$$\frac{I_0}{K-I_0} = Ye^{r(0)}$$

$$Y = \frac{I_0}{K-I_0}$$

Replacing value of Y into (eq3),

$$I(t) = \frac{\left(\frac{I_0}{K-I_0}\right)Ke^{rt}}{1 + \left(\frac{I_0}{K-I_0}\right)e^{rt}}$$

Multiplying both the numerator and denominator by $(K - I_0)$,

$$\frac{\left(\frac{I_0}{K-I_0}\right)Ke^{rt}}{1 + \left(\frac{I_0}{K-I_0}\right)e^{rt}} \times \frac{(K-I_0)}{(K-I_0)} = \frac{I_0Ke^{rt}}{(K-I_0) + I_0e^{rt}} = \frac{I_0Ke^{rt}}{K + I_0(e^{rt} - 1)}$$

$$\therefore I(t) = \frac{I_0Ke^{rt}}{K + I_0(e^{rt} - 1)} \quad (eq4)$$

Applying Integrated Logistic Growth Equation to COVID-19 Context:

$$K = \text{Plateau Value of real world data} = 330415$$

$$I_0 = 4372 \text{ cases (Total cumulative case on 2 Aug 2021)}$$

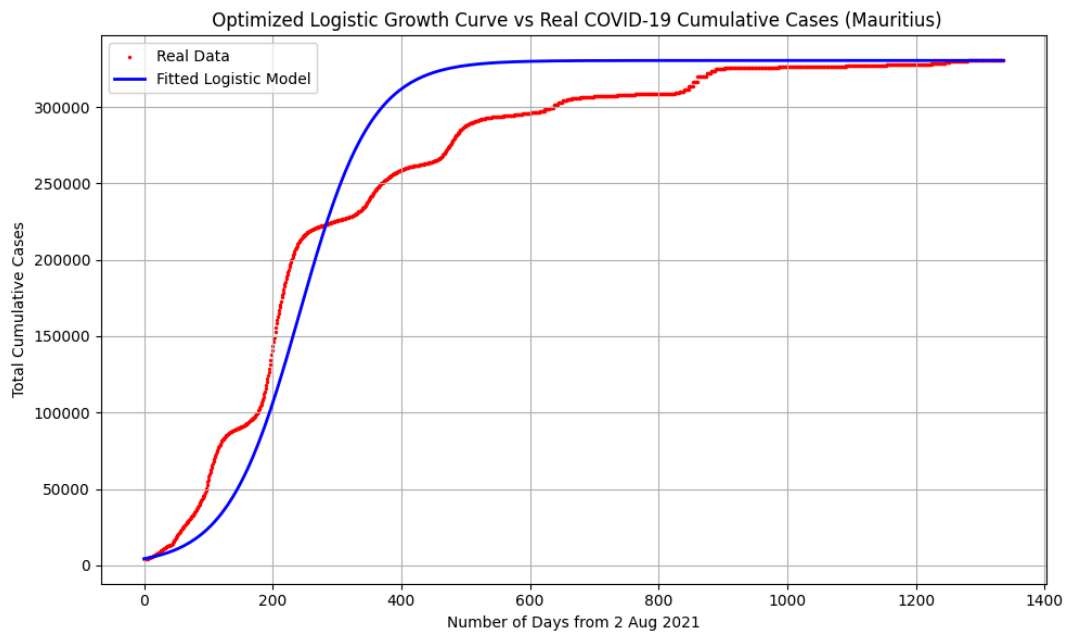
Rearranging (eq4) to make r the subject of formula,

$$r = \frac{\ln\left(\frac{I_2(K-I_0)}{I_0(K-I_2)}\right)}{t}$$

R is calculated from two points before the cumulative cases reach the plateau value. The first point is $I_0 = 4372$ cases and second point is at 24 Apr 2022 where $I_2 = 220551$ cases and the time period between the two points is $t = 266$ days. Hence, the predicted value of r is:

$$r_{\text{predicted}} = \frac{\ln\left(\frac{220551(330415-4372)}{4372(330415-220551)}\right)}{266} = 0.01883 \approx 0.0188 \text{ (3 s.f.)}$$

The values of K and I_0 will remain fixed whilst the value of r may vary as the Logistic Growth Model will be optimised to closely fit the real-world data set using the `curve_fit` function from `scipy.optimize` module. The Logistic Growth Model is programmed using Pycharm (Appendix) which gives the following output and graph:



The above graph illustrates that there is significant disparity between the Logistic Growth Model and the real-world data, which is reinforced by the high RMSE and % Error Margin. The optimised growth rate(r) calculated by the program is 0.0178 which is 0.001 less than the predicted growth rate; the values are quite close. The % Error Margin(10.84%) obtained will be used in the comparison of the 3 models.

```
Logistic Growth Model Summary

Growth Rate (r):           0.0178
Initial Cumulative Cases (P0): 4,372
Final Cumulative Cases (K): 330,415
RMSE:                      28197.48 cases
% Error Margin : 10.84%

Process finished with exit code 0
```

SIR Model

Predicting the values of parameters and initial populations:

The values of $S(0)$, $I(0)$ and $R(0)$ are calculated as shown below:

$$I(0) = \text{active cases on 2 Aug 2021} = 267 \text{ cases}$$

Number of infected + Number of recovered or dead = Total Cumulative cases on 2 Aug 2021

$$I(0) + R(0) = 4372 \text{ cases}$$

$$\therefore R(0) = 4372 - 267 = 4105 \text{ cases}$$

$$S(0) = \text{Mauritian population} - \text{Total Cumulative cases}$$

$$S(0) = 1\,270\,000 - 4372 = 1\,265\,628 \text{ cases}$$

The basic reproduction number indicates how many people an infected person contacts on average while they are infected which was 9.66 in Mauritius; quite high given the fact that the population is relatively small compared to other countries. A person's infectious period is estimated to last on average 10 days. Hence the infection rate (β) would be the likelihood that a susceptible person will contract the virus after coming in contact with an infected individual during the latter's infectious period in a fixed population.

$$\beta_{\text{predicted}} = \frac{\left(\frac{\text{Basic Reproduction Number}}{\text{Duration of infectious period}}\right)}{\text{Population}} = \frac{\left(\frac{9.66}{10}\right)}{1.27 \text{ million}} = \frac{0.966}{(1.27 \times 10^6)} = 7.61 \times 10^{-7} \text{ (3s.f.)}$$

If a person remains infected for an average of 2 weeks, then the likelihood of them recovering on any given day is $\frac{1}{14}$. Hence,

$$\gamma_{\text{predicted}} = \frac{1}{14} = 0.0714$$

Solving SIR Equations using Euler's Method:

Thus, replacing the values of the parameters in the SIR's differential equations:

$$\frac{dS}{dt} = -(7.606 \times 10^{-7})SI \quad \frac{dI}{dt} = (7.606 \times 10^{-7})SI - 0.0714I \quad \frac{dR}{dt} = 0.0714I$$

From Euler's method for numerical integration, these equations would be implemented:

$$x_i = x_{i-1} + h \quad y_i = y_{i-1} + h \times f(x_{i-1}, y_{i-1})$$

Where, h is the step size for each increment (for this application it is 1 day) and x_i is the x-coordinate of the i th point (i.e. x_i 's equation will be ignored; $i = t$ for application) and y_i is the y-coordinate of the i th point (i.e. the value of a specific population on day x_i). Implementing the equation for y_i for the 3 respective populations,

$$S_t = S_{(t-1)} + \frac{dS}{dt} \Rightarrow S_t = S_{(t-1)} - (7.61 \times 10^{-7})S_{(t-1)}I_{(t-1)}$$

$$I_t = I_{(t-1)} + \frac{dI}{dt} \Rightarrow I_t = I_{(t-1)} + (7.61 \times 10^{-7})S_{(t-1)}I_{(t-1)} - 0.0714I_{(t-1)}$$

$$R_t = R_{(t-1)} + \frac{dR}{dt} \Rightarrow R_t = R_{(t-1)} + 0.0714 \times I_{(t-1)}$$

The table below shows an example of the start of solving the differential equations:

t	$S_{(t-1)}$	$I_{(t-1)}$	$R_{(t-1)}$	$\frac{dS}{dt}$	$\frac{dI}{dt}$	$\frac{dR}{dt}$	S_t	I_t	R_t	Total Cumulative cases = $I_t + R_t$
1	1 265 628	267	4105	- 257.03	237.97	19.06	1 265 370.97	504.97	4124.06	4629.03
...										

The whole calculation was automated by writing a function in Python for up until day 1336:

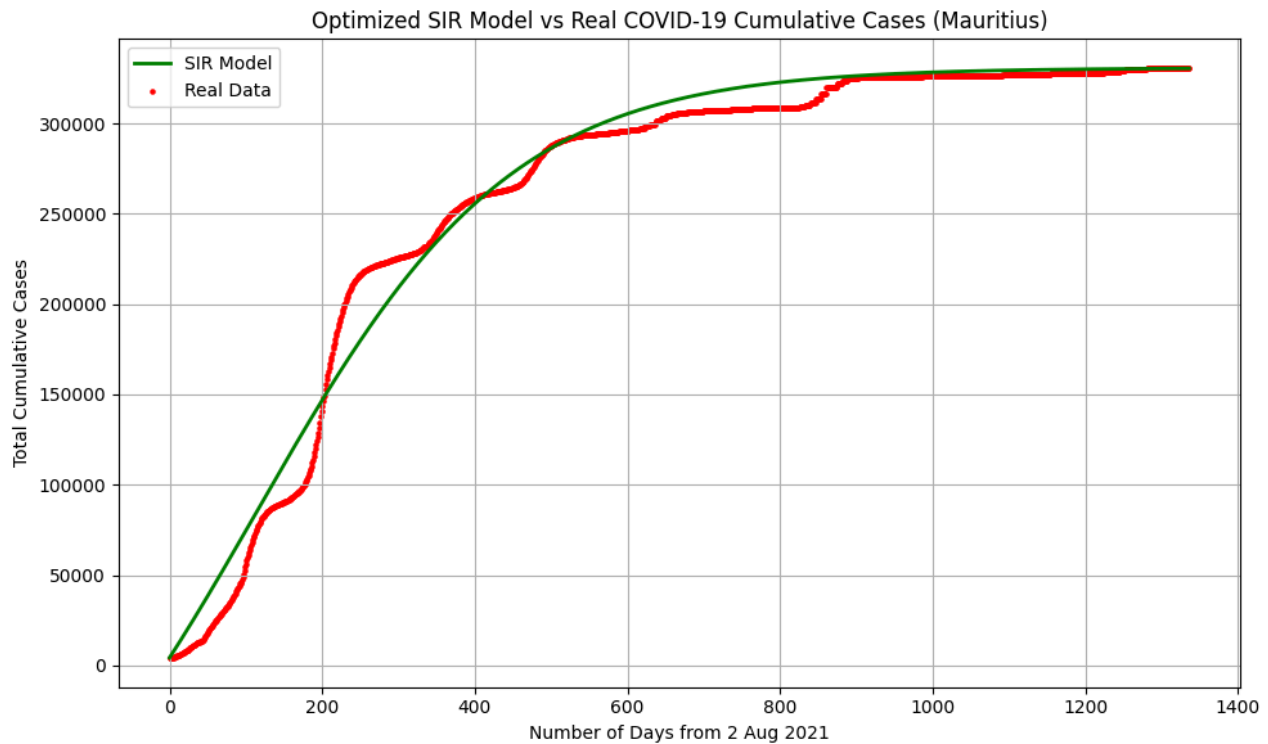
```

35 # SIR Model using Euler's Method
36
37 def sir_model(beta, gamma, I0, R0, days):
38     S = np.zeros(days)
39     I = np.zeros(days)
40     R = np.zeros(days)
41     cumulative = np.zeros(days)
42
43     S[0] = N - I0 - R0
44     I[0] = I0
45     R[0] = R0
46     cumulative[0] = I0 + R0
47
48     for i in range(1, days):
49         dS = -beta * S[i-1] * I[i-1] / N
50         dI = beta * S[i-1] * I[i-1] / N - gamma * I[i-1]
51         dR = gamma * I[i-1]
52
53         S[i] = S[i-1] + dS
54         I[i] = I[i-1] + dI
55         R[i] = R[i-1] + dR
56
57         cumulative[i] = I[i] + R[i]

```

Applying SIR Model to COVID-19 Context:

The initial and final total cumulative cases ($I(0) + R(0)$) will remain fixed whilst the values of $I(0)$, $R(0)$, β and γ will vary as the SIR Model will be optimised to closely fit the real-world data set using the minimize function from scipy.optimize module. Pycharm (Appendix) was used to program the SIR Model, producing the results and graph shown below:



The above graph illustrates that the SIR Model is quite close to the real-world data, which is reinforced by the low RMSE and % Error Margin. The optimised $I(0)$ and $R(0)$ calculated respectively have a big disparity between their predicted population; optimised $I(0)$ is 91 cases less whilst optimised $R(0)$ is 91 cases more. Nevertheless, the program managed to make the initial and final cumulative cases constant. It can also be seen that the model deviates from the predicted values of the infection rate and recovery rate. Hence, the calculated % Error Margin and the SIR Model's optimised parameters and populations will be used in the comparison between the 3 models.

```
Prediction for Initial Populations
Initial Infected ( $I_0$ ):      267
Initial Recovered ( $R_0$ ):    4104.884195
Initial Susceptible ( $S_0$ ): 1265628.115805
SIR Model Summary
Optimized  $\beta$ : 2.677183857068941e-07
Optimized  $\gamma$ : 0.3368
Optimized  $I_0$ : 176
Optimized  $R_0$ : 4195
Initial Cumulative Cases: 4,372
Final Model Cases:      330,415
Final Actual Cases:     330,415
RMSE:                   12268.31 cases
% Error Margin:         4.72%
Process finished with exit code 0
```

SEIQR Model

Predicting the values of parameters and initial populations:

For this model, $I(0)$ will be the same as in the SIR Model, $I(0) = 267$ cases. Since, it is difficult to find the values for $E(0)$ and $Q(0)$, $E(0)$ will be assumed to be 100 cases whilst $Q(0)$ would be 50 cases, these predicted values will change when the SEIQR Model is optimised to fit the real-world data curve.

$$\begin{array}{ccccccc} \text{Number of} & + & \text{Number of} & + & \text{Number of} & = & \text{Total Cumulative cases on 2 Aug 2021} \\ \text{infected} & & \text{quarantined} & & \text{recovered or dead} & & \end{array}$$

$$I(0) + Q(0) + R(0) = 4372 \text{ cases}$$

$$\therefore R(0) = 4372 - (267 + 50) = 4055 \text{ cases}$$

$$S(0) = \text{Mauritian population} - (\text{Total Cumulative cases} + \text{Number of exposed})$$

$$S(0) = 1\,270\,000 - (4372 + 100) = 1\,265\,528 \text{ cases}$$

Compared to the SIR Model having only 2 parameters, the SEIQR Model has 7 parameters. The infection rate β will also be the same as in SIR Model ($\beta_{\text{predicted}} = 7.61 \times 10^{-7}$).

The incubation period for COVID-19 is assumed to last for 5 days on average, hence the likelihood of them getting symptoms is $\frac{1}{5}$, hence the predicted incubation rate is $\sigma_{\text{predicted}} = 0.2$.

Since the recovery rate γ is broken into $\gamma_E, \gamma_I, \gamma_Q$, these parameters will arbitrarily be assigned the same value as the predicted recovery rate in the SIR Model, ($\gamma_{\text{predicted}} = 0.0714$).

It can be predicted that the values of these 3 parameters will vary between one another when the SEIQR Model is optimised. As it is difficult to calculate the quarantine rates δ_E, δ_I , they will arbitrarily be assigned the value of 0.1.

The predicted values of all the parameters and the initial populations will definitely change when the model is optimised whilst keeping the initial cumulative cases fixed.

Solving SEIQR Equations using Euler's Method:

Thus, replacing the values of the parameters in the SIR's differential equations:

$$\frac{dS}{dt} = - (7.606 \times 10^{-7})SI$$

$$\frac{dE}{dt} = (7.606 \times 10^{-7})SI - (0.2 + 0.0714 + 0.0714)E \Rightarrow \frac{dE}{dt} = (7.606 \times 10^{-7})SI - 0.3428E$$

$$\frac{dI}{dt} = 0.2E - (0.0714 + 0.1)I \Rightarrow \frac{dI}{dt} = 0.2E - 0.1714I$$

$$\frac{dQ}{dt} = 0.1E + 0.1I - 0.0714Q$$

$$\frac{dR}{dt} = 0.0714E + 0.0714I + 0.0714Q \Rightarrow \frac{dR}{dt} = 0.0714(E + I + Q)$$

From Euler's method for numerical integration, implementing the equation for y_i for the 5 respective populations,

$$S_t = S_{(t-1)} + \frac{dS}{dt} \Rightarrow S_t = S_{(t-1)} - (7.61 \times 10^{-7})S_{(t-1)}I_{(t-1)}$$

$$E_t = E_{(t-1)} + \frac{dE}{dt} \Rightarrow E_t = E_{(t-1)} + (7.606 \times 10^{-7})S_{(t-1)}I_{(t-1)} - 0.3428E_{(t-1)}$$

$$I_t = I_{(t-1)} + \frac{dI}{dt} \Rightarrow I_t = I_{(t-1)} + 0.2E_{(t-1)} - 0.1714I_{(t-1)}$$

$$Q_t = Q_{(t-1)} + \frac{dQ}{dt} \Rightarrow Q_t = Q_{(t-1)} + 0.1E_{(t-1)} + 0.1I_{(t-1)} - 0.0714Q_{(t-1)}$$

$$R_t = R_{(t-1)} + \frac{dR}{dt} \Rightarrow R_t = R_{(t-1)} + 0.0714(E_{(t-1)} + I_{(t-1)} + Q_{(t-1)})$$

Simplifying the aforesaid equations,

$$S_t = S_{(t-1)} - (7.61 \times 10^{-7})S_{(t-1)}I_{(t-1)}$$

$$E_t = 0.6572E_{(t-1)} + (7.606 \times 10^{-7})S_{(t-1)}I_{(t-1)}$$

$$I_t = 0.8286I_{(t-1)} + 0.2E_{(t-1)}$$

$$Q_t = 0.9286Q_{(t-1)} + 0.1E_{(t-1)} + 0.1I_{(t-1)}$$

$$R_t = R_{(t-1)} + 0.0714(E_{(t-1)} + I_{(t-1)} + Q_{(t-1)})$$

The tables below shows an example of the start of solving the differential equations:

t	$S_{(t-1)}$	$E_{(t-1)}$	$\frac{dS}{dt}$	$\frac{dE}{dt}$	S_t	E_t
1	1 265 528	100	- 257.03	222.75	1 265 370.97	322.75
...						

t	$I_{(t-1)}$	$Q_{(t-1)}$	$R_{(t-1)}$	$\frac{dI}{dt}$	$\frac{dQ}{dt}$	$\frac{dR}{dt}$	I_t	Q_t	R_t	Total Cumulative cases = $I_t + Q_t + R_t$
1	267	50	4055	- 25.76	33.13	29.77	241.24	83.13	4084.77	4409.14
...										

The whole calculation was automated by writing a function in Python for up until day 1336:

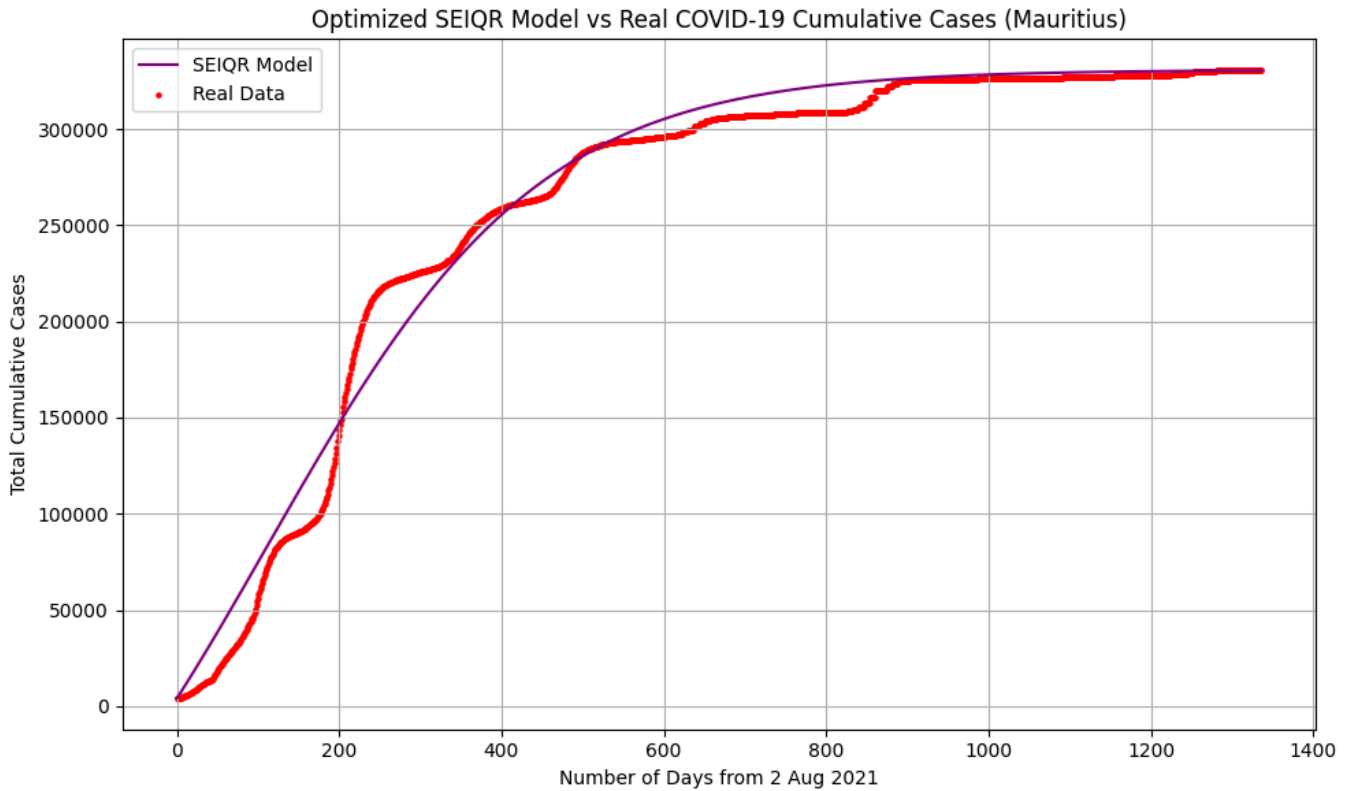
```

41
42     S[0], E[0], I[0], Q[0], R[0] = S0, E0, I0, Q0, R0_guess
43     cumulative[0] = initial_cumulative
44
45     for i in range(1, days):
46         dSdt = -beta * S[i-1] * I[i-1] / N
47         dEdt = beta * S[i-1] * I[i-1] / N - (sigma + gammaE + deltaE) * E[i-1]
48         dIdt = sigma * E[i-1] - (gammaI + deltaI) * I[i-1]
49         dQdt = deltaE * E[i-1] + deltaI * I[i-1] - gammaQ * Q[i-1]
50         dRdt = gammaE * E[i-1] + gammaI * I[i-1] + gammaQ * Q[i-1]
51
52         S[i] = max(S[i-1] + dSdt, 0)
53         E[i] = max(E[i-1] + dEdt, 0)
54         I[i] = max(I[i-1] + dIdt, 0)
55         Q[i] = max(Q[i-1] + dQdt, 0)
56         R[i] = max(R[i-1] + dRdt, 0)
57
58         cumulative[i] = I[i] + Q[i] + R[i]

```

Applying SEIQR Model to COVID-19 Context:

The initial and final total cumulative cases ($I(0) + Q(0) + R(0)$) will remain fixed whilst the values of $E(0), I(0), Q(0), \beta, \sigma, \gamma_E, \gamma_I, \gamma_Q, \delta_E$ and δ_I will vary as the SEIQR Model will be optimised to closely fit the real-world data set using the minimize function from scipy.optimize module. The SEIQR Model was programmed using Pycharm (Appendix), giving the following graph and results:



The graph above indicates how closely the SEIQR Model matches the actual data, which is further supported by the low RMSE and % Error Margin. The optimised $E(0)$, $I(0)$ and $Q(0)$ calculated are very close to their predicted population. The program also managed to make the initial and final cumulative cases constant. The optimised β is very close to the predicted infection rate as well as γ_I being the closest recovery rate to its predicted value. However, there is an obvious difference in the optimised incubation rate and a small inaccuracy as the optimised γ_E was higher than the optimised γ_I whilst the likelihood of an exposed person to recover without getting infected is less than an infected person recovering after gaining symptoms. Therefore, the determined percentage error margin as well as the optimised parameters and populations of the SEIQR Model will be utilised to compare the three models.

Initial Population Guesses

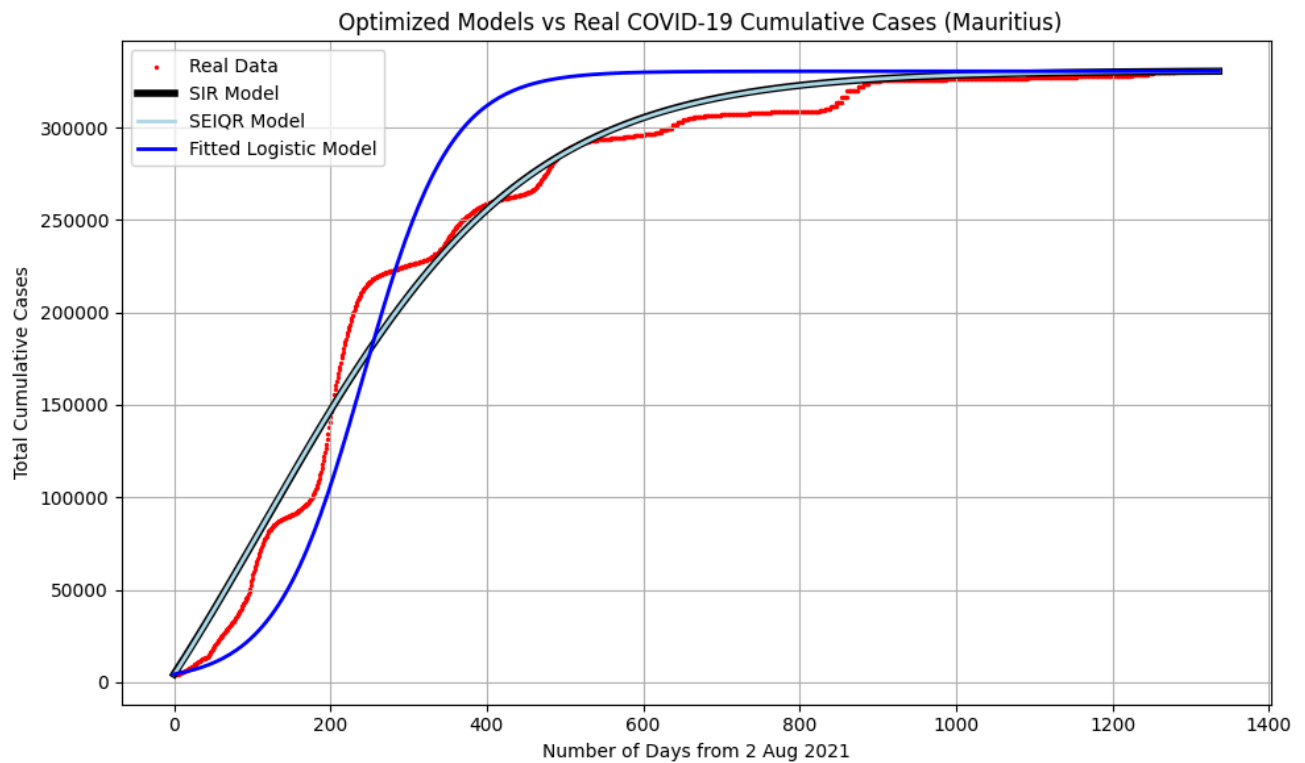
```
Initial Exposed ( $E_0$ ):      100
Initial Infected ( $I_0$ ):     267
Initial Quarantined ( $Q_0$ ):  50
Initial Recovered ( $R_0$ ):    4054.8841949999996
Initial Susceptible ( $S_0$ ): 1265528.115805
```

SEIQR Model Evaluation Summary

```
Optimized  $\beta$ : 7.633433172755002e-07
Optimized  $\sigma$ : 0.0563
Optimized  $\gamma_E$ : 0.1358
Optimized  $\gamma_I$ : 0.0716
Optimized  $\gamma_Q$ : 0.1083
Optimized  $\delta E$ : 0.1493
Optimized  $\delta I$ : 0.0851
Initial Cumulative Cases ( $P_0$ ): 4,372
Optimized Initial Infected ( $I_0$ ): 262
Optimized Initial Quarantined ( $Q_0$ ): 47
Optimized Initial Exposed ( $E_0$ ): 98
Initial Recovered ( $R_0$ ): 4054.8841949999996
Final Cumulative Cases (model): 330,415
Final Cumulative Cases (actual): 330,415
RMSE: 12166.32 cases
% Error (RMSE / mean actual): 4.68%
```

```
Process finished with exit code 0
```

Conclusion



As mentioned previously, this IA's aim is to determine which of the aforementioned models is the most accurate in predicting the propagation of COVID-19 in Mauritius.

Starting in ascending order of accuracy, the Logistic Growth Model was the least accurate with a RMSE of 28 197.48 *cases* and the highest % error margin of 10.84% . Hence, its inaccuracy is due to the model being too simplistic as it does not account the different stages of infection, incubation periods, policy changes and behavioural reactions whilst assuming a constant rate of spread and a natural slowing based on population constraints.

The SIR Model demonstrated a high degree of accuracy, surpassing the Logistic Growth Model but falling short of the SEIQR Model; with an RMSE of 12 268.31 *cases* and a small % error margin of 4.72%. Despite its % error margin being very small, there is a huge difference in its optimised infection and recovery rates and the ones that were predicted. On top of that, the optimised $I(0) = 176$ *cases* is far from the 267 active cases on 2 Aug 2021. This inaccuracy could be the result of not taking the exposed and quarantined populations into account.

In fulfilling the aim of the IA, the SEIQR Model is the most accurate model in predicting the propagation of COVID-19 in Mauritius during the selected time period; with the lowest RMSE of 12 166.32 *cases* and % error margin of 4.68 %. Its high accuracy is accentuated by the optimised

parameters and populations being close to their predicted values. Due to the fact that the selected time period span over several years, the optimised incubation rate was smaller than its predicted value (incubation period being higher $20\text{ days} > 5\text{ days}$) as the model may incorporate other external factors such as policy effects and testing & reporting delays into one single value. In Spite of the % error margins of SEIQR and SIR being around the same range, the SEIQR Model is more accurate as its $I(0) = 262\text{ cases}$ which is close to the 267 active cases. The SEIQR Model's highest accuracy would be due to its additional populations, enabling it to record transmission delays and the aftermath of isolation measures; thus being the most suited model amongst the others to capture the real-world propagation of COVID-19 in Mauritius.

Evaluation

Limitations:

The accuracy of the cumulative case predictions over time may be lowered by the assumptions made by all the 3 models, such as constant parameter values, which does not account for shifting real-world circumstances. As mentioned earlier, the large selected time period could result in external factors incorporated in the values of parameters. Moreover, the completeness and quality of the data used to fit each model may have a significant impact on cumulative case prediction accuracy.

Strengths:

One of the strengths of this IA is the comparative modelling approach which enabled a clearer understanding of each model's assumptions and limitations and fulfilled the aim of this IA. Moreover, the 3 models were programmed using Python which minimised human error and allowed for the models to be optimised to fit the real-world data graph. Furthermore, the application of the models in the COVID-19 Context in Mauritius demonstrated an understanding of the discrepancy between theoretical models and real-world epidemiological behavior.

Extension to the IA:

An extension to this investigation, would be to make the parameters time-dependent for the SIR and SEIQR Models. This would take into account actual changes that occurred during the time period, lockdowns and new variants of the virus. This will enable the models to be more dynamic and more accurate in depicting the evolving nature of propagation over time using piecewise functions; being more complex.

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Appendices

Logistic Growth Model Program

```
import pandas as pd

import numpy as np

import matplotlib.pyplot as plt

from scipy.optimize import curve_fit

from sklearn.metrics import mean_squared_error

# Loading Real-world data

df = pd.read_csv("MauritiusCOVID.csv")

df = df[df['Entity'] == 'Mauritius'].reset_index(drop=True)

real_cases = df['Total confirmed cases of COVID-19 per million people'].to_numpy() * 1.27

real_days = np.arange(len(real_cases)) # Days since start

# Fixing initial and final total cumulative case

P0_fixed = real_cases[0]

K_fixed = real_cases[-1]

# Logistic model with only r as a parameter

def logistic_fixed_P0_K(t, r):

    return (P0_fixed * K_fixed * np.exp(r * t)) / ((K_fixed - P0_fixed) + P0_fixed * np.exp(r * t))

# Initial guess for r

initial_guess = [0.01883]

# Fitting the model to the Real-world data

popt, _ = curve_fit(logistic_fixed_P0_K, real_days, real_cases, p0=initial_guess)

r_fit = pop[0]

print(f" Fitted growth rate r: {r_fit:.4f}")

# Generating model values
```

```

t_model = np.linspace(0, real_days[-1], 300)

model_cases = logistic_fixed_P0_K(t_model, r_fit)

# Calculating predictions for actual days for RMSE

model_fit_to_actual = logistic_fixed_P0_K(real_days, r_fit)

rmse = np.sqrt(mean_squared_error(real_cases, model_fit_to_actual))

# Calculating percentage error margin from RMSE

mean_actual = np.mean(real_cases)

percent_error = (rmse / mean_actual) * 100

# Outputting the information

print("\n Logistic Growth Model Summary")

print("")

print(f" Growth Rate (r):           {r_fit:.4f}")

print(f" Initial Cumulative Cases (P0): {P0_fixed:,.0f}")

print(f" Final Cumulative Cases (K):   {K_fixed:,.0f}")

print(f" RMSE:                        {rmse:.2f} cases")

print(f" % Error Margin : {percent_error:.2f}%")

# Plotting both graphs

plt.figure(figsize=(10, 6))

plt.scatter(real_days, real_cases, color='red', label='Real Data', s=2)

plt.plot(t_model, model_cases, color='blue', label='Fitted Logistic Model', linewidth=2)

plt.title(" Optimized Logistic Growth Curve vs Real COVID-19 Cumulative Cases (Mauritius)")

plt.xlabel("Number of Days from 2 Aug 2021")

plt.ylabel("Total Cumulative Cases")

plt.grid(True)

plt.legend()

plt.tight_layout()

plt.show()

```

SIR Model Program

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

from sklearn.metrics import mean_squared_error

from scipy.optimize import minimize

# Loading Real-world data

df = pd.read_csv("MauritiusCOVID.csv")

df = df[df['Entity'] == 'Mauritius'].reset_index(drop=True)

real_cases = df['Total confirmed cases of COVID-19 per million people'].to_numpy() * 1.27

t = np.arange(len(real_cases))

days = len(t)

N = 1_270_000 # Total population

initial_cumulative = real_cases[0]

target_final = real_cases[-1]

# Initialising population guess

I0_guess = 267

R0_guess = initial_cumulative - I0_guess

S0_guess = N - I0_guess - R0_guess

print("Prediction for Initial Populations")

print("")

print(f"Initial Infected (I0): {I0_guess}")

print(f"Initial Recovered (R0): {R0_guess}")

print(f"Initial Susceptible (S0): {S0_guess}")

# SIR Model using Euler's Method

def sir_model(beta, gamma, I0, R0, days):

    S = np.zeros(days)

    I = np.zeros(days)

    R = np.zeros(days)
```

```

cumulative = np.zeros(days)

S[0] = N - I0 - R0

I[0] = I0

R[0] = R0

cumulative[0] = I0 + R0

for i in range(1, days):

    dS = -beta * S[i-1] * I[i-1] / N

    dI = beta * S[i-1] * I[i-1] / N - gamma * I[i-1]

    dR = gamma * I[i-1]

    S[i] = S[i-1] + dS

    I[i] = I[i-1] + dI

    R[i] = R[i-1] + dR

    cumulative[i] = I[i] + R[i]

# Scaling so that start and end match the real data

if cumulative[-1] != cumulative[0]:

    scaling_factor = (target_final - cumulative[0]) / (cumulative[-1] - cumulative[0])

    cumulative[1:] = cumulative[0] + (cumulative[1:] - cumulative[0]) * scaling_factor

return cumulative

# Objective function for optimization

def objective(params, t, real_cases):

    beta, gamma, I0 = params

    R0 = initial_cumulative - I0

    model = sir_model(beta, gamma, I0, R0, len(t))

```

```

if np.any(np.isnan(model)) or np.any(np.isinf(model)):
    return np.inf

return np.sqrt(mean_squared_error(real_cases, model))

# Optimization

initial_guess = [7.606299*(10**-7), 0.0714, I0_guess] # [beta, gamma, I0]

bounds = [
    (0, 0.5), # beta
    (0, 0.5), # gamma
    (0, initial_cumulative) # I0 must be ≤ initial cases
]

result = minimize(objective, initial_guess, args=(t, real_cases), bounds=bounds)

beta_opt, gamma_opt, I0_opt = result.x

R0_opt = initial_cumulative - I0_opt

# Final model and evaluation

final_model = sir_model(beta_opt, gamma_opt, I0_opt, R0_opt, days)

rmse = np.sqrt(mean_squared_error(real_cases, final_model))

percent_error = (rmse / np.mean(real_cases)) * 100

# Output results

print("SIR Model Summary")

print("")

print(f"Optimized β: {beta_opt/N}")

print(f"Optimized γ: {gamma_opt:.4f}")

print(f"Optimized I₀: {I0_opt:.0f}")

print(f"Optimized R₀: {R0_opt:.0f}")

print(f"Initial Cumulative Cases: {initial_cumulative:,.0f}")

print(f"Final Model Cases:      {final_model[-1]:,.0f}")

print(f"Final Actual Cases:      {target_final:,.0f}")

```

```

print(f"RMSE:                {rmse:.2f} cases")

print(f"% Error Margin:      {percent_error:.2f}%")

# Plotting

plt.figure(figsize=(10, 6))

plt.plot(t, final_model, label="SIR Model", color='green', linewidth=2)

plt.scatter(t, real_cases, label="Real Data", color='red', s=5)

plt.title("Optimized SIR Model vs Real COVID-19 Cumulative Cases (Mauritius)")

plt.xlabel("Number of Days from 2 Aug 2021")

plt.ylabel("Total Cumulative Cases")

plt.legend()

plt.grid(True)

plt.tight_layout()

plt.show()

```

SEIQR Model Program

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

from sklearn.metrics import mean_squared_error

from scipy.optimize import minimize

# Loading Real-world data

df = pd.read_csv("MauritiusCOVID.csv")

df = df[df['Entity'] == 'Mauritius'].reset_index(drop=True)

real_cases = df['Total confirmed cases of COVID-19 per million people'].to_numpy() * 1.27

t = np.arange(len(real_cases))

N = 1_270_000 # Total population

initial_cumulative = real_cases[0]

target_final = real_cases[-1]

# Initial population guesses

I0_guess = 267 # initial infected guess

Q0_guess = 50 # initial quarantined guess

R0_guess = initial_cumulative - I0_guess - Q0_guess

E0_guess = 100 # initial exposed guess

S0 = N - E0_guess - I0_guess - Q0_guess - R0_guess

print("\n Initial Population Guesses")

print("")

print(f"Initial Exposed ( $E_0$ ): {E0_guess}")

print(f"Initial Infected ( $I_0$ ): {I0_guess}")

print(f"Initial Quarantined ( $Q_0$ ): {Q0_guess}")

print(f"Initial Recovered ( $R_0$ ): {R0_guess}")

print(f"Initial Susceptible ( $S_0$ ): {S0}")
```

```
# SEIQR Model using Euler's Method
```

```
def seiqr_model(beta, sigma, gammaE, gammal, gammaQ, deltaE, deltal, I0, Q0, E0, days):  
    S, E, I, Q, R = np.zeros(days), np.zeros(days), np.zeros(days), np.zeros(days),  
np.zeros(days)  
  
    cumulative = np.zeros(days)  
  
    S[0], E[0], I[0], Q[0], R[0] = S0, E0, I0, Q0, R0_guess  
    cumulative[0] = initial_cumulative  
  
    for i in range(1, days):  
        dSdt = -beta * S[i-1] * I[i-1] / N  
        dEdt = beta * S[i-1] * I[i-1] / N - (sigma + gammaE + deltaE) * E[i-1]  
        dIdt = sigma * E[i-1] - (gammal + deltal) * I[i-1]  
        dQdt = deltaE * E[i-1] + deltal * I[i-1] - gammaQ * Q[i-1]  
        dRdt = gammaE * E[i-1] + gammal * I[i-1] + gammaQ * Q[i-1]  
  
        S[i] = max(S[i-1] + dSdt, 0)  
        E[i] = max(E[i-1] + dEdt, 0)  
        I[i] = max(I[i-1] + dIdt, 0)  
        Q[i] = max(Q[i-1] + dQdt, 0)  
        R[i] = max(R[i-1] + dRdt, 0)  
  
        cumulative[i] = I[i] + Q[i] + R[i]  
  
    # Scale only from day 1 onward so both start and end match  
    if cumulative[-1] != cumulative[0]:  
        scaling_factor = (target_final - cumulative[0]) / (cumulative[-1] - cumulative[0])  
        cumulative[1:] = cumulative[0] + (cumulative[1:] - cumulative[0]) * scaling_factor
```

```

return cumulative

# Objective function

def objective(params, t, real_cases):
    try:
        beta, sigma, gammaE, gammal, gammaQ, deltaE, deltal, I0, Q0, E0 = params
        model_cases = seiqr_model(beta, sigma, gammaE, gammal, gammaQ, deltaE, deltal, I0,
Q0, E0, len(t))
        if np.any(np.isnan(model_cases)) or np.any(np.isinf(model_cases)):
            return np.inf
        return np.sqrt(mean_squared_error(real_cases, model_cases))
    except Exception:
        return np.inf

# Optimization

initial_guess = [7.606299*(10**-7), 0.2, 0.0714, 0.0714,0.0714, 0.1, 0.1, I0_guess, Q0_guess,
E0_guess]
bounds = [
    (0, 1), # beta
    (0, 1), # sigma
    (0, 1), # gammaE
    (0, 1), # gammal
    (0, 1), # gammaQ
    (0, 1), # deltaE
    (0, 1), # deltal
    (0, 1000), # I0 (initial infected)
    (0, 1000), # Q0 (initial quarantined)
    (0, 1000) # E0 (initial exposed)
]

```

```

# Minimize the objective function
result = minimize(objective, initial_guess, args=(t, real_cases), bounds=bounds)

beta, sigma, gammaE, gammal, gammaQ, deltaE, deltal, I0_opt, Q0_opt, E0_opt = result.x

# Final model and evaluation
best_model = seiqr_model(beta, sigma, gammaE, gammal, gammaQ, deltaE, deltal, I0_opt,
Q0_opt, E0_opt, len(t))

rmse = np.sqrt(mean_squared_error(real_cases, best_model))

percent_error = (rmse / np.mean(real_cases)) * 100

# Output results
print("\n SEIQR Model Evaluation Summary")
print("")
print(f" Optimized  $\beta$ : {beta/N}")
print(f" Optimized  $\sigma$ : {sigma:.4f}")
print(f" Optimized  $\gamma_E$ : {gammaE:.4f}")
print(f" Optimized  $\gamma_I$ : {gammal:.4f}")
print(f" Optimized  $\gamma_Q$ : {gammaQ:.4f}")
print(f" Optimized  $\delta_E$ : {deltaE:.4f}")
print(f" Optimized  $\delta_I$ : {deltal:.4f}")
print(f" Initial Cumulative Cases ( $P_0$ ): {initial_cumulative:,.0f}")
print(f" Optimized Initial Infected ( $I_0$ ): {I0_opt:.0f}")
print(f" Optimized Initial Quarantined ( $Q_0$ ): {Q0_opt:.0f}")
print(f" Optimized Initial Exposed ( $E_0$ ): {E0_opt:.0f}")
print(f" Initial Recovered ( $R_0$ ): {R0_guess}")
print(f" Final Cumulative Cases (model): {best_model[-1]:,.0f}")
print(f" Final Cumulative Cases (actual): {target_final:,.0f}")
print(f" RMSE: {rmse:.2f} cases")
print(f" % Error (RMSE / mean actual): {percent_error:.2f}%")

```

```
# Plotting

plt.figure(figsize=(10, 6))

plt.plot(t, best_model, label="SEIQR Model", color='purple')

plt.scatter(t, real_cases, label="Real Data", color='red', s=5)

plt.title("Optimized SEIQR Model vs Real COVID-19 Cumulative Cases (Mauritius)")

plt.xlabel("Number of Days from 2 Aug 2021")

plt.ylabel("Total Cumulative Cases")

plt.legend()

plt.grid(True)

plt.tight_layout()

plt.show()
```